SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project 7: Shared Resource Scheduling Engine (Rooms & Fleet Vehicles)

Version: 1.0
Tech Stack: MySQL (XAMPP), React.js (Frontend), Node.js/Express (Backend), Socket.io (Real-time Availability Updates), node-cron (Reminders), fullcalendar / react-big-calendar (UI), Nodemailer (Notifications)

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1. Executive Summary

A centralized, real-time resource booking platform that enables staff to discover, reserve, and manage shared organizational resources—meeting rooms, training labs, conference halls, and fleet vehicles. The system provides a visual calendar interface with color-coded availability, conflict prevention, automated conflict resolution suggestions, and a robust check-in/out workflow. It dramatically reduces double-booking incidents and improves resource utilization across the organization.

2. Problem Statement & Objectives

- **Problem:** Organizations rely on manual booking methods (WhatsApp, sticky notes, Excel spreadsheets) leading to double-booked rooms, idle vehicles, frustrated staff, and wasted productivity. There is no visibility into resource availability or usage patterns.
- **Objectives:** (1) Digitize resource discovery and booking with real-time availability. (2) Prevent double-booking with conflict detection. (3) Provide intelligent "next available" suggestions. (4) Implement a check-in/check-out workflow with auto-release for no-shows. (5) Enable recurring bookings and waitlisting. (6) Generate utilization reports for capacity planning.

3. Scope

- **In-Scope:** Web application; Resource management (Rooms, Vehicles, Labs, Equipment); Visual calendar (daily/weekly/monthly); Conflict prevention; Recurring bookings; Check-in/Check-out; Auto-cancellation for no-shows; Waitlist; Email notifications; Basic usage analytics.
- **Out-of-Scope:** Integration with Outlook/Google Calendar (V2); IoT-based room occupancy sensors (V2); GPS tracking for vehicles (V2); Mobile native apps (PWA in V2); Payment/charging for external users.

4. Stakeholders

- **End-Users:** Staff (Bookers), Resource Managers (Custodians), Department Heads (Approvers, if required).
- **Internal:** Super Admin, System Admin.
- **Technical:** Developers, QA, DBA.

5. User Roles & Permissions (RBAC Matrix)

Module / Action	Super Admin	Resource Manager	Department Head	Staff (Regular)	Auditor
Manage System Settings	✔	✘	✘	✘	✘
Create/Edit/Delete Resources	✔	✔	✘	✘	✘
View All Resources & Calendars	✔	✔	✔ (Dept only)	✔	✔
Book a Resource (Self)	✔	✔	✔	✔	✘
Book on Behalf of Others	✔	✔	✘	✘	✘
Approve/Reject Booking (if required)	✔	✔ (Own resources)	✔ (Dept only)	✘	✘
Cancel/Reschedule Booking (Self)	✔	✔ (Any)	✔ (Dept only)	✔ (Own only)	✘
Check-In/Check-Out Resource	✔	✔	✘	✔ (Self only)	✘
View Utilization Reports	✔	✔	✔ (Dept only)	✘	✔
View Full Audit Logs	✔	✘	✘	✘	✔

6. Functional Requirements (FR) - Module Wise

Module 1: Resource Management (FR-001 to FR-008)

FR-001: Admin/Resource Manager creates a new Resource with:

- Name, Description, Type (Meeting Room, Conference Hall, Training Lab, Vehicle, Equipment).
- Capacity (number of people for rooms/vehicles).
- Location/Building/Room Number.
- Features/Amenities (e.g., Projector, Whiteboard, Video Conferencing, WiFi).
- Operating Hours (default: 8:00 AM – 6:00 PM, Sunday–Thursday).
- Booking Lead Time (e.g., min 1 hour before booking).
- Max Booking Duration (e.g., 4 hours).
- Default Booking Duration (e.g., 60 minutes).
- Is Approval Required? (Yes/No – configurable per resource).
- Is Check-In Required? (Yes/No – if yes, auto-release on no-show).

- **FR-002:** Resources can have Maintenance Schedules (pre-booked blocks).
- **FR-003:** Resources can be temporarily Disabled (e.g., under renovation, vehicle in service).
- **FR-004:** Resources can be categorized (e.g., "Small Meeting Rooms", "Large Conference Halls", "Sedan Vehicles", "SUVs").
- **FR-005:** Each resource has a unique Resource ID (e.g., RM-101, VH-202).
- **FR-006:** Admin can upload photos/images of the resource.
- **FR-007:** Resources can be tagged with departments (e.g., "IT Dept Only", "Finance Dept Only", or "All Departments").
- **FR-008:** System supports Resource Groups (e.g., "Book any available room in Building A").

Module 2: Booking Calendar & Discovery (FR-009 to FR-013)

FR-009: Staff logs in → Navigates to "Book Resource".

FR-010: Calendar view options:

- **Daily View:** Hour-by-hour grid for a specific resource.
- **Weekly View:** All resources in a category with color-coded availability.
- **Monthly View:** Overview of busy/available days.

FR-011: Color-coding:

- **Green:** Available.
- **Red/Orange:** Booked.
- **Yellow:** Pending Approval.
- **Gray:** Maintenance/Disabled.

- **FR-012:** Hover over a slot shows: Resource Name, Current Booking (if any), Capacity, Features.
- **FR-013:** Advanced filters: Capacity (min seats), Features (Projector/WiFi/Video Call), Location, Resource Type.

Module 3: Booking Engine (FR-014 to FR-023) - Core

FR-014: Staff selects a resource → Clicks on an available time slot → Opens booking form.

FR-015: Booking form fields:

- Title/Agenda (mandatory, max 200 chars).
- Start Date & Time (auto-filled from clicked slot).
- End Date & Time (auto-calculated; staff can adjust).
- Recurring Options (Daily, Weekly, Monthly – with end date).
- Number of Attendees (optional, for capacity checks).
- Special Requirements (text, optional).

FR-016: System validates:

- Staff has permission to book the resource.
- Resource is available for the selected time.
- Booking duration does not exceed max allowed.
- Booking time is within operating hours.
- Capacity is not exceeded.

FR-017: Conflict detection: If the selected slot overlaps with an existing booking, system:

- Blocks the booking.
- Displays "Conflict Detected" message.
- Suggests Next Available Times (e.g., "Next available slot is 2:00 PM").

- **FR-018:** If resource requires approval, booking status = Pending Approval → Approvers notified.
- **FR-019:** If no approval required, booking status = Confirmed immediately.
- **FR-020:** Recurring bookings: System creates individual bookings for each occurrence (max 52 weeks).
- **FR-021:** Staff can book on behalf of another staff member (if role allows).
- **FR-022:** System generates a unique Booking Reference Number (e.g., BK-2026-0042).
- **FR-023:** Upon confirmation, staff receives email with booking details.

Module 4: Conflict Resolution & Smart Suggestions (FR-024 to FR-026)

FR-024: When a conflict occurs, system displays:

- Next Available Slot (same resource).
- Alternative Resources (same type/category) that are available at the selected time.

- **FR-025:** System can suggest Adjacent Slots (e.g., "Book from 11:00 AM instead of 10:30 AM to avoid conflict").
- **FR-026:** Waitlist: If no alternative is available, staff can join a Waitlist. If a slot opens (due to cancellation), the first waitlisted user is notified with a time-limited offer to book.

Module 5: Check-In/Check-Out & No-Show Management (FR-027 to FR-031)

FR-027: For resources where check-in is required, staff must check-in within a Grace Period (configurable, default 15 minutes) of the booking start time.

FR-028: Check-In can be done via:

- Web interface (click "Check-In" on booking).
- QR Code scan at the resource location (optional V2).

FR-029: If no check-in occurs within the grace period:

- Booking is auto-cancelled (No-Show).
- Resource becomes available for others.
- Staff is flagged with a "No-Show" penalty (configurable: after 3 no-shows, booking privileges restricted).

- **FR-030:** Check-Out (optional): Staff clicks "Check-Out" when leaving; system logs actual end time.
- **FR-031:** Feedback: After check-out, staff can rate the resource (1-5 stars) and provide comments.

Module 6: Approval Workflow (FR-032 to FR-036)

FR-032: Resources configured with Requires Approval = Yes trigger an approval workflow.

FR-033: Approver (Resource Manager or Department Head) receives a notification with booking details.

FR-034: Approver can:

- **Approve:** Booking status → Confirmed. Staff notified.
- **Reject:** Booking status → Rejected. Mandatory reason required. Staff notified.
- **Hold:** Booking status → Pending. Comments added (e.g., "Need budget approval").

- **FR-035:** If no action is taken within SLA (configurable, default 24 hours), system sends a reminder to the approver.
- **FR-036:** After 48 hours (configurable), if still pending, system auto-escalates to the approver's supervisor.

Module 7: Cancellation & Rescheduling (FR-037 to FR-040)

- **FR-037:** Staff can cancel their booking up to 2 hours before the start time (configurable) without penalty.
- **FR-038:** Late cancellations (< 2 hours) are flagged and count towards a penalty score.
- **FR-039:** Staff can reschedule to another available time slot (subject to availability).
- **FR-040:** Resource Managers can cancel or reschedule any booking (with a mandatory reason, sent to the staff).

Module 8: Notifications & Reminders (FR-041 to FR-045)

FR-041: Confirmation Email: Sent immediately after booking is confirmed.

FR-042: Reminders:

- 24 hours before: Email with booking details and location.
- 1 hour before: Email reminder (or SMS – optional).
- 15 minutes before: In-app notification (if check-in is required).

- **FR-043:** Conflict/Change Notifications: Staff is notified if a booking is cancelled/changed by an Admin.
- **FR-044:** Waitlist Notifications: When a slot opens, waitlisted users are notified with a 24-hour offer window to claim the slot.
- **FR-045:** No-Show Summary: Resource Manager receives a weekly summary of no-shows (Friday 5:00 PM).

Module 9: Reports & Analytics (FR-046 to FR-050)

FR-046: Dashboard KPI Cards:

- Total Resources, Total Bookings (Today/This Week), Utilization Rate (%), No-Show Rate, Pending Approvals.

FR-047: Utilization Reports:

- Per Resource: % of available time booked.
- Per Category: Meeting Rooms vs. Vehicles vs. Labs.
- Peak Hours/Days: When resources are most heavily used.

- **FR-048:** No-Show Reports: List of no-shows by staff, department, and resource.
- **FR-049:** Department Usage: Which departments are using which resources (for cost allocation).
- **FR-050:** Reports exportable as CSV/PDF.

7. Non-Functional Requirements (NFR)

- **NFR-001 (Performance):** Calendar loads in < 2 seconds for 50+ resources. Conflict detection < 200ms.
- **NFR-002 (Real-Time):** Socket.io for live availability updates when bookings are created/cancelled.
- **NFR-003 (Security):** JWT authentication; staff can only view resources they have permission to book.
- **NFR-004 (Reliability):** 99.5% uptime. Automated daily backups.
- **NFR-005 (Usability):** Fully responsive (mobile-first) for on-the-go check-in and booking.

8. Business Rules (BR)

- **BR-001:** A staff member cannot book more than 2 meetings simultaneously (overlap rule).
- **BR-002:** No-show 3 times within 30 days → Booking privileges suspended for 7 days (configurable).
- **BR-003:** Maximum booking duration is 4 hours per session (configurable per resource).
- **BR-004:** Recurring bookings cannot exceed 12 weeks into the future.
- **BR-005:** Vehicles cannot be booked for intra-building use only (require destination validation – V2).
- **BR-006:** Emergency bookings (flag marked) can override approval requirements.

9. User Stories (US)

- **US-01:** As a Staff Member, I want to find an available meeting room near my office with a projector, book it for 2 hours, and receive a confirmation email.
- **US-02:** As a Resource Manager, I want to approve/reject vehicle booking requests and mark vehicles as "Out of Service" when they are in maintenance.
- **US-03:** As a Department Head, I want to see my department's resource utilization to optimize costs.
- **US-04:** As an Admin, I want to auto-cancel bookings where staff don't check-in, so resources are not wasted.
- **US-05:** As a Staff Member, I want to be waitlisted for a fully booked room and get notified if a slot opens.

10. Use Cases (High-Level)

- **UC-01: Book a Meeting Room** (Staff → System): Search for available rooms → Select room + time → Fill form → Confirm (if no approval needed) → Receive confirmation.
- **UC-02: Check-In to a Booking** (Staff → System): Go to the room → Open app → Click "Check-In" → System logs timestamp → Booking status updated.
- **UC-03: Approve a Vehicle Booking** (Manager → System): View pending request → Review details → Approve/Reject with reason → System notifies staff.
- **UC-04: Manage Resource Calendar** (Manager → System): View full-month calendar → Block time for maintenance → System prevents bookings during blocked time.

11. Activity & Workflow (End-to-End: Staff Journey)

1. Staff logs in → Navigates to "Book Resource".
2. Staff selects "Meeting Room" category → Views available rooms.
3. Staff selects a room → Views its calendar (daily view).
4. Staff clicks an available time slot → Booking form appears.
5. Staff fills in Title, Duration, Attendees → Clicks "Book".
6. System checks availability → Confirms (if no approval required).
7. System sends confirmation email with QR code (if check-in required).
8. Staff arrives at the room at the scheduled time → Opens app → Clicks "Check-In".
9. System logs check-in → Room is marked as "Occupied" on the live board.
10. Staff finishes meeting → Clicks "Check-Out" (optional).
11. System sends a feedback prompt.

12. System Modules

- **Resource Management Module** (CRUD, categorization, maintenance scheduling).
- **Calendar & Discovery Module** (Visual calendar, filters, search).
- **Booking Engine Module** (Conflict detection, smart suggestions, waitlist).
- **Approval Workflow Module** (Pending approvals, notifications, escalation).
- **Check-In/Out Module** (QR scanning, grace period enforcement, no-show handling).
- **Notification Module** (Email, in-app, reminders).
- **Reporting & Analytics Module** (Utilization, no-shows, peak times).
- **Audit & Logging Module** (Immutable logs).
- **User & Role Management Module** (Authentication, RBAC).

13. Screen-by-Screen Checklist (UI)

Screen	Key Elements
Dashboard	KPI cards (Bookings Today, Utilization, Pending Approvals, No-Shows). Upcoming bookings list, Quick "Book Resource" button.
Resource Discovery	Search bar, Category tabs, Filter sidebar (Capacity, Features, Location). Resource cards with availability badge.
Resource Calendar	Toggle (Daily/Weekly/Monthly), Clickable time slots, Color-coded availability, Quick book button.

Screen	Key Elements
Booking Form	Resource name & image, Title input, Start/End time pickers, Recurring options, Attendees count, Special requirements text, Submit/Cancel buttons.
Conflict Modal	Conflict warning, Suggested next available slot(s), Alternative resources list, Waitlist option.
My Bookings	Upcoming bookings table, Past bookings history, Action buttons (Cancel/Reschedule/Check-In).
Approval Dashboard (Manager)	Pending requests list, Approve/Reject buttons, Reason text box, Booking detail view.
Check-In Page	Today's bookings list, "Check-In" button (active 15 min before start), QR code display.
Admin Settings	Resource types, Operating hours, Grace period, No-show penalty rules, Approval configuration.

14. Database Tables (ERD - Core)

Table Name	Key Fields
resources	id, resource_uuid (unique), name, type (room/vehicle/lab/equipment), capacity, location, features (JSON), operating_hours_start, operating_hours_end, max_duration_minutes, default_duration_minutes, requires_approval (boolean), requires_checkin (boolean), is_active, department_restriction (FK departments/nullable)
resource_availability_exceptions	id, resource_id (FK), date, start_time, end_time, reason (Maintenance/OutOfService/Other)
bookings	id, booking_ref (unique), resource_id (FK), booked_by (FK users), title, start_datetime, end_datetime, is_recurring (boolean), recurrence_rule (JSON, if recurring), attendees (int, optional), special_requirements (text), status (pending/confirmed/in-progress/completed/cancelled/no-show/rejected), created_at, updated_at
booking_occurrences	id, booking_id (FK), occurrence_start, occurrence_end, status (confirmed/completed/cancelled/no-show) – used for recurring bookings
check_ins	id, booking_id (FK), checked_in_at, checked_out_at, check_in_method (web/QR/manual)
approvals	id, booking_id (FK), approver_id (FK users), status (pending/approved/rejected), reason, approved_at
waitlist	id, resource_id (FK), requested_by (FK users), desired_start, desired_end, status (waiting/offered/accepted/expired), created_at
no_show_penalties	id, user_id (FK), booking_id (FK), penalty_count (int), penalty_start, penalty_end (if suspended)
feedback	id, booking_id (FK), rating (1-5), comment, submitted_at
audit_logs	id, user_id (FK), action (CREATE/UPDATE/CANCEL/APPROVE/REJECT/CHECKIN/CHECKOUT), booking_id (FK), details (JSON), timestamp
users	Same as previous projects with role enum.

15. API Checklist (RESTful)

Group	Endpoint	Method	Description
Resources	/api/resources	GET	List resources (filterable by type/location)
	/api/resources/:id	GET	Get resource details
	/api/resources/:id/availability	GET	Get availability for a date range
	/api/resources	POST	Create resource (Admin)
Bookings	/api/bookings	POST	Create booking
	/api/bookings/my	GET	Get logged-in staff's bookings
	/api/bookings/:id	GET	Get booking details
	/api/bookings/:id/cancel	POST	Cancel booking

Group	Endpoint	Method	Description
	/api/bookings/:id/reschedule	PUT	Reschedule booking
Check-In	/api/bookings/:id/checkin	POST	Check-in to booking
	/api/bookings/:id/checkout	POST	Check-out of booking
Approvals	/api/approvals/pending	GET	Get pending approvals (Manager)
	/api/approvals/:id/action	POST	Approve/Reject booking
Conflict	/api/resources/:id/next-available	GET	Get next available slot
	/api/resources/alternatives	POST	Find alternative resources
Waitlist	/api/waitlist	POST	Join waitlist
	/api/waitlist/offer/:id	POST	Accept waitlist offer
Reports	/api/reports/utilization	GET	Utilization report
	/api/reports/no-shows	GET	No-show report
Admin	/api/admin/resources/:id/block	POST	Block resource for maintenance
	/api/admin/penalties	GET	View all penalties

16. Validation Rules

- **Booking Time:** Start time must be in the future, at least min_lead_time ahead.
- **Booking Duration:** Must be ≤ max_duration_minutes per resource.
- **Resource Capacity:** Attendees count ≤ resource capacity (if provided).
- **Recurring End Date:** Must be ≤ 12 weeks from start date.
- **Check-In Window:** Can check-in from -15 minutes to +15 minutes of start time.

17. Security Requirements

- **Resource Isolation:** Staff can only book resources they have permission for (department restrictions enforced).
- **JWT Authentication:** Required for all endpoints.
- **Rate Limiting:** 20 booking requests per minute per user.
- **SQL Injection:** Parameterized queries.
- **HTTPS:** Enforced.

18. Notifications

Email (Nodemailer/SendGrid):

- Booking Confirmation (with calendar .ics attachment).
- 24-hour reminder.
- 1-hour reminder.
- Approval Required notification (to Approver).
- Approval/Rejection notification (to Staff).
- Cancellation notification (if Admin cancels).
- Waitlist offer notification.
- No-Show penalty notice.

In-App (Toast/Socket.io):

- Booking created/cancelled updates (live).
- Check-in reminders.

19. Reports & Analytics

- **Resource Utilization Report:** % utilization per resource (daily/weekly/monthly).
- **No-Show Report:** Staff-wise, department-wise, resource-wise.
- **Peak Usage Report:** Identify busiest days and times.
- **Department Usage Report:** Cost allocation basis (if implemented).

- **Capacity Planning Report:** Identify over-utilized vs. under-utilized resources.

20. Audit Logs

- **Logged Actions:** Resource creation, booking creation/cancellation/rescheduling, check-in/out, approval/rejection, waitlist joins/offers, penalty enforcement.
- **Retention:** Logs retained for 3 years. Auto-archived annually.

21. Testing Checklist (QA)

Type	Key Test Cases
Functional	Staff books room → Confirmed → Arrives → Check-in → Leaves → Check-out → Feedback submitted.
Conflict	Two staff try to book same room at same time → Second gets conflict warning → Suggests next available.
Recurring	Book a weekly meeting for 8 weeks → All 8 occurrences created.
No-Show	Staff doesn't check-in within 15 min → Auto-cancelled → Penalty recorded.
Approval	Resource requires approval → Manager approves → Status updated → Staff notified.
Performance	Calendar loads 50+ resources in < 2 seconds. Real-time updates reflect instantly.

22. Deployment Checklist (Production)

- Set NODE_ENV=production and env variables.
- Run migrations with indexes on resource_id, start_datetime, status, booked_by.
- Configure Nginx to proxy WebSocket connections for real-time updates.
- Install Redis (for Socket.io scaling and rate limiting).
- Setup cron jobs: 0 8 * * * – Send daily reminders; 0 * * * * – Auto-cancel no-shows (every hour); 0 17 * * 5 – Weekly utilization report to Admin.
- PM2 setup with --max-memory-restart 1G.
- SSL certificate installed.
- Test email delivery.

23. Acceptance Criteria (Go/No-Go)

- Staff can find an available room, book it, and receive a confirmation email within 60 seconds.
- The system prevents double-booking of the same resource at the same time.
- A no-show booking is auto-cancelled and the resource becomes available again.
- Staff can check-in via the web interface, and the system logs the timestamp.
- Resource Manager can approve/reject bookings and block resources for maintenance.
- The utilization dashboard correctly displays the % of booked time for each resource.
- Real-time updates (via WebSockets) reflect booking changes instantly for all users.